

BENJAMIN E. JORDAN

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LinkedIn | Personal Website

SKILLS

Languages: Python, Java, TypeScript, C++, C **Tools:** Git, Docker, AWS, Runpod, Linux, HuggingFace, LaTeX
Libraries: NumPy, Spark, Pandas, PyTorch, Pytest, Scikit-learn, Boto3, Faiss, DDP/FSDP, Optuna, Angular, Selenium

EXPERIENCE

Software Engineer @ Amazon Oct 2024 – Present
Measurement, Ad Tech, & Data Science New York, NY

- Productionized brand-awareness model pipeline using AWS SageMaker, which processes 500+ million rows of data daily, producing model predictions and aggregating survey responses to generate lift metrics for brand lift reports
- Designed and delivered a workflow to automate the model training, evaluation, and approval process on a weekly cadence
- Enabled shadowing of experimental models in a production environment
- Designs for the items above became ML ops templates/references for multiple other large ML projects around the org
- Worked extensively on large-scale spark ETL jobs (in AWS EMR and AWS Glue)
- Contributed to other components written in Python, Java, Typescript, and Scala
- Collaborated frequently with surveys science team to implement new methodologies for measuring advertising performance

Machine Learning Engineer @ Northrop Grumman Feb 2024 – Oct 2024
Autonomous Intelligence & Robotics Lab Denver, CO

- Enabled accurate, real-time object detection by distilling GroundingDINO into YOLOv10, decreasing inference latency on Jetson Orin from 0.6s to 0.08s
- Researched techniques for implementing and evaluating retrieval-augmented generation (RAG) systems, and led the technical side of a project focused on classified PDF retrieval and question answering
- Implemented a RAG system utilizing ANN search and locally running vision-language models for PDF Q&A
- Wrote a survey on techniques for mitigating a lack of labeled training data: active learning, semi-supervised learning, knowledge distillation, and synthetic data generation

Machine Learning Engineer @ KLA May 2023 – Aug 2023
Internship Ann Arbor, MI

- Independently worked on semiconductor defect detection algorithms
- Fine-tuned and quantized vision transformers for efficient and robust defect classification with limited data
- Presented project during poster board session and was invited to give a virtual talk on transformers and transfer learning

Software Engineer @ Carestream May 2022 – Aug 2022
Internship Rochester, NY

- Developed and maintained backend functionality in Carestream's ImageView x-ray software

PROJECTS

Entropy Audio | Website | Project Blog Jan 2024 – Present
Personal project focused on novel sound design tools for composers

- Curated a multi-terabyte audio sample dataset using web scraping, open-source data, synthetic description generation and augmentation, and manual labeling
- Implemented policy gradient reinforcement learning algorithms for diffusion models, using pretrained reward models to better optimize for downstream objectives. Increased average CLAP score by $\approx 20\%$ using GRPO.
- Implemented the loss from the Diffusion-DPO paper and trained audio generation model on a preference dataset collected through the EntropyAudio website UI
- Created code packages for training, model code, evaluation metrics, and data curation - drawing inspiration from open source codebases such as stable-audio-tools (Stability AI), audiobox-aesthetics (Meta FAIR), audiocraft (Meta FAIR)
- Implemented a web app for text-based audio generation that incorporates a preference data flywheel into the user workflow
- Implemented a backend system in AWS and RunPod to support inference, data storage, and other application logic

EDUCATION

Cornell University 2022 – 2023
Master's in Computer Science, Concentration in Machine Learning, 3.76 GPA Ithaca, NY

Rochester Institute of Technology 2018 – 2022
Bachelor's in Computer Science, Minor in Music, Presidential Merit Scholarship, 3.96 GPA during 2nd half of degree Rochester, NY

NOTABLE COURSEWORK

Machine Learning: Large-Scale Machine Learning, Machine Learning Theory, Reinforcement Learning, Machine Learning Hardware and Systems, Computer Vision, Numerical Linear Algebra, Natural Language Processing